João Pedro Sousa Ferreira

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Summary

Assistant Professor and researcher at the University of Porto (FEUP) specializing in Artificial Intelligence and Computational Mechanics. Pioneered deep learning methods for automatic medical image analysis (endoscopy, cholangioscopy), translating research into patented technologies and a University of Porto spin-off. Former Principal Investigator of FCT-funded research projects, with over 5M€ in secured funding; consulting in structural and image analysis generating over €150k. 90+ journal publications, h-index 23.

Professional Experience

Chief Data Scientist, DigestAID — University of Porto spin-off — Feb 2026 – present
Porto, Portugal
Leads the company's AI and data strategy for medical imaging: research roadmap, regulated ML engineering (SaMD) and the data-science team.

- Inventor on 8 patent families for deep-learning detection of gastrointestinal lesions (granted in PT, US, EP, GB)
- Patented capsule-endoscopy tool in routine use at a partner hospital, supporting clinicians with faster lesion detection

Head Data Scientist, DigestAID — University of Porto spin-off — Jun 2021 – Jan 2026
Porto, Portugal
Took AI for medical imaging from research prototypes to Software as a Medical Device in clinical use.

- Built and led a team of 8 data scientists; owned the technical architecture behind $10M in funding
- Four AI products taken through FDA/EMA submissions under ISO 13485 (SaMD), delivered to regulatory approval

Researcher, INEGI — Institute of Science and Innovation in Mechanical and Industrial Engineering (LAETA), University of Porto — Sep 2014 – Sep 2024
Porto, Portugal
Computational biomechanics and AI research; PhD researcher (2014–2021), then Principal Investigator of national and industry-funded projects.

- Principal Investigator: FCT PTDC/EME-APL/1342/2020 (€249k), NVIDIA Applied Research Accelerator hardware grant, FCT advanced-computing allocation
- Team member in six further FCT/EU projects on soft-tissue biomechanics and medical AI
- Consultancy in structural and image analysis (>€150k), including the PROMEDON medical-device biocompatibility study (2020–2021)

Visiting Researcher (Fulbright award), Cleveland Clinic / Case Western Reserve University — Jan 2019 – Jul 2019
Cleveland, OH, USA
Research stay on the biomechanics of pelvic-floor tissues and smooth-muscle cells; Fulbright incentive award.

Consultancy and R&D Contracts

DigestAID / INEGI — Jun 2021 – present
Porto, Portugal
Analysis and integration of medical image databases for classification problems (ongoing project). FEUP – Faculty of Engineering, University of Porto.

PROMEDON — Nov 2020 – May 2021
Porto, Portugal
Assessment of structural integrity and biocompatibility for market integration of a medical device (completed and delivered project). INEGI – Institute of Science and Innovation in Mechanical and Industrial Engineering.

DigestAID — Mar 2021
Porto, Portugal
Knowledge transfer to the DigestAID spin-off resulting from provisional patent applications (completed processes). FEUP – Faculty of Engineering, University of Porto.

Teaching Career

Invited Assistant Professor, Department of Mechanical Engineering, Faculty of Engineering, University of Porto — Aug 2020 – present
since 19 August 2020

- Technical Drawing, Mechanical Design Drawing, Numerical Analysis, Calculus, Linear Algebra and Analytic Geometry, Computer Programming

Invited Teaching Assistant, Department of Mechanical Engineering, Faculty of Engineering, University of Porto — Sep 2017 – Jul 2020
between 1 September 2017 and 30 July 2020
- Mechanics II, Structural Mechanics I, Computer-Aided Design and Manufacturing, Computer-Aided Drawing

Tutor, Faculty of Engineering, University of Porto — Oct 2010 – Jun 2012
between 1 October 2010 and 30 June 2012
- Mathematics, Physics, and Programming Help Desk

Academic Qualifications

University of Porto — Mechanical Engineering (PhD) — Mar 2021
Porto (OPO)
PhD in Mechanical Engineering, unanimously approved (Cum Laude)
- Thesis: A cell-based approach to early detect female pelvic organ prolapse

Faculty of Engineering, University of Porto — Mechanical Engineering – Mechanical Design profile (MSc) — Sep 2014
Porto (OPO)
MSc in Mechanical Engineering – Mechanical Design profile
- Thesis: Elasto-plastic analysis of structures using an isogeometric formulation

Skills

Machine Learning & AI: deep learning for medical image analysis, mechanics-informed neural networks, model deployment to clinical settings

Computational Mechanics: continuum mechanics, soft-tissue biomechanics, finite element analysis (Abaqus UMAT)

Programming: Python (scientific ML stack), MATLAB, Fortran (UMAT)

Research leadership: PI on national/European grants, PhD/MSc supervision, clinical-academic collaborations (Cleveland Clinic, Hospital de São João)

Patents

Detection and differentiation of gynecologic lesions in colposcopy — Aug 2023
Ferreira, João P S, others
WO2025211973A1 · Patent inventor (application). Family: PT119365A (pub. Oct 2025). Assignee: Gynoaid / U.Porto – FEUP · link

Method for classification of anorectal manometry exams data — Aug 2022
Mascarenhas Saraiva, Miguel, **Ferreira, João P S**, others
PT118356 · Patent inventor (granted). Family: WO2024112220A1, EP4623450A1 (pending). Assignee: U.Porto – FEUP · link

Automatic detection and differentiation/classification of the esophagus, stomach, small bowel and colon lesions in device-assisted enteroscopy using a convolutional neuronal network — Aug 2021
Ferreira, João P S, others
PT117392 · Patent inventor (granted); EP4384979B1 granted Dec 2024 (ES3012550T3). Family: WO2023018344A1. Assignee: U.Porto – FEUP · link

Automatic detection and differentiation of pancreatic cystic lesions in endoscopic ultrasonography — Aug 2021
Ferreira, João P S, others
PT117391 · Patent inventor (registered). Family: WO2023018343A1, US20240354949A1 and DE112022003919T5 pending. Assignee: U.Porto – FEUP · link

Automatic detection and differentiation of biliary lesions in cholangioscopy images — Feb 2021
Ferreira, João P S, others
PT117086 · Patent inventor (granted); EP4298593B1 granted Jul 2025, GB2618003B granted Oct 2025. Family: WO2022182263A1, US20240135540A1 pending. Assignee: U.Porto – FEUP · link

Automatic detection of erosions and ulcers in Crohn's capsule endoscopy — Nov 2020
Ferreira, João P S, others
PT116896 · Patent inventor (registered); US 12,573,031 B2 granted Mar 2026. Family: WO2022108466A1, GB2615477A. Assignee: U.Porto – FEUP · link

Automatic detection of colon lesions and blood in colon capsule endoscopy — Nov 2020
Ferreira, João P S, others
PT116895 · Patent inventor (registered); US 12,488,459 B2 granted Dec 2025. Family: WO2022108465A1, GB2615476A. Assignee: U.Porto – FEUP · link

Automatic detection and differentiation of small bowel lesions in capsule endoscopy — Nov 2020
Ferreira, João P S, others
PT116894 · Patent inventor (granted). Family: WO2022108464A1, EP4252183A1, US20240013377A1, GB2615274A, CA3199518A1, CN116830148A. Assignee: U.Porto – FEUP · link

Full inventor profile on Google Patents

Awards and Distinctions

Distinction Posters, Digestive Disease Week (DDW) 2022–2024
Poster distinctions at Digestive Disease Week conferences

Auxiliary Award for Top 3 Abstracts, American College of Gastroenterology (ACG 2022) Oct 2022
Deep Learning for Automatic Diagnosis and Morphologic Characterization of Malignant Biliary Strictures Using Digital Cholangioscopy: A Multicentric Study

Selected Scientific Publication, IACOBUS (500 EUR) Apr 2021
Characterization of hyperelastic and damage behavior of tendons

Travel Award, Basic Sciences Symposium, American Urology Association (AUA 2019) May 2019
Chicago, Illinois
Stiffness of vaginal smooth muscle cells: Loxl1 knockout vs wildtype (NIH Award number 1R13DK121565)

Fulbright Incentive Award for research work in the USA Jun 2018
Cleveland, OH
Cleveland Clinic / Case Western Reserve University — research stay 1 January to 30 July 2019

Best Abstract, MIPS V Annual Meeting — Mediterranean Incontinence and Pelvic Floor Society (MIPS 2018) May 2018
Rome, Italy
Accurate in silico-simulation of the ovine bladder based on mechanical and histological data

Selected Publications

90+ papers in international journals and 240+ total scientific outputs, h-index 23 (1600+ citations). Full list: ORCID · Google Scholar